RNN-based Deep Learning for One-hour ahead Load Forecasting

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***Abstract*— This paper is focusing on one hour-ahead forecasting on Power Load, using Recurrent Neural Network based scheme. This study only uses the generated data of Kookmin University 's Load, so it required a considerable number of resources for forecasting. Multi-scaled RNN model was proposed for the Load Forecasting, which is suitable for both short term and long term memory.**

Keywords— Machine Learning, Output Load Forecasting, Recurrent Neural Network.

I. INTRODUCTION (*HEADING 1*)

Load Forecasting is attracting a lot of attention recently, as it plays an important role in electrical power utilities. Load Forecasting supplies useful decision-making basic for power management, especially renewable energy, such as Photovoltaic and Wind Power, fuel supplies scheduling, maintenance scheduling… [1]. These predictions are both important in local energy management and the modern energy market. Load tendency is affected by many elements and hard to get accuracy forecast results. Some Load Forecasting methods are researched, of which Neural Networks are often adopted as high accuracy and flexible when dealing with different input data. In the Neural Network and Deep Learning method, Recurrent Neural Network is considered as a robust solution for serial data prediction, which is very suitable for Load Forecasting [2]. In this paper, we discuss about One-hour ahead Forecasting in power load system, using RNN method.

The remain of this paper is as follows: Section II introduces the Recurrent Neural Network and the overall architecture of the model. Section III explains our proposed Recurrent Neural Network implementation and evaluation. Lastly, we will conclude the paper in Section IV.

II. APPLY OF MACHINE LEARNING TECHNIQUE

A. Recurrent Neural Network

A recurrent neural network (RNN) is a class of artificial neural networks where the output of one unit is the input of another unit, to form a sequence of graph, so that it has an internal state (memory) and exhibit temporal dynamic behaviors. Because of the special structure, RNN has the ability to remember the sequences of inputs, reflect in the internal state of nodes. RNN is a robust approach for series pattern recognition, such as handwriting recognition or speech recognition, or predict the time-series data [3].

The RNN has the characteristic of temporal dynamic behavior, which can have additional stored state that operations under direct control of the neural network. The unit's temporal storage state can be replaced by another network, if that incorporates time delays or has feedback loops. Such temporal storage states can be gated state or gated memory and also be gated recurrent units.

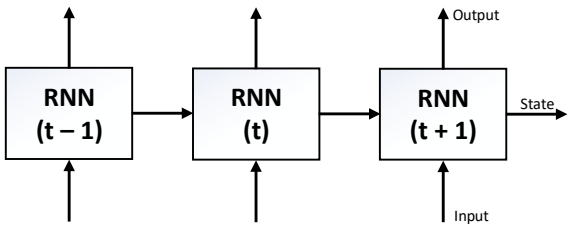


Figure 1: Proposed RNN structure

Compare to other ANN model, the proposed RNN-based deep learning can process longer duration data as input and can reflect time-series-properties due to the unique memory cell and recursive neural structures. Our study focuses on analyzing and evaluating the proposed forecasting model, with the different hyper-parameters for the RNN techniques. The proposed model expected to extract the unique long term characteristic of the input data. The longer the duration of input data from output prediction, the harder to predict future pattern. In contracts, with the larger number of data considered in the prediction, the recent data become blurred and vanished, so that the procedure of the forecasting model cannot be done. We can avoid such problems by design RNN model to receive a shorter duration such as 6h, instead of 1 week (24h x 7). The length of considered data become smaller, increasing the relationship with the immediately preceding data, but reduce the ability to extract the long-term pattern [4].

B. Multi-Scale RNN-based Deep Learning Architecture

The overall framework of the Output Load Forecasting is illustrated in Figure 2. The time-series Output Load data are divided into two different time-scales of data (6-hour duration for the short scale and 1-week duration for long scale) and fed into RNNs. The Extracted features layer is also divided into two parts with the ratios of 32 and 168, which then oriented to the output-specific features of Output Load data. The extracted features are then concatenated and utilized for the

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prediction of the next hour after feedforwarding through the fully connected layer [4].

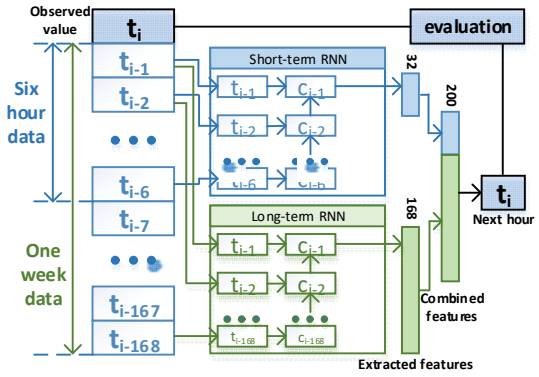


Figure 2: Proposed multi-scale RNN-based framework for Output Load Forecasting [4]

The proposed architecture of Short-term RNN and Long-term RNN have two main functions that can be jointly considered, which is reemphasizing short-term data by appropriately reflecting unexpected and unpatternizable intrinsic temporal factors and maintaining the overall accuracy of RNN model based on the long-term data. The Combined feature layer combines the unique features of short-term and long-term RNN networks, which can re-emphasize unexpected and unpatternizable intrinsic temporal factors with weighting intrinsically and the forecasting ability of long-term one, is the key factor of the multi-scale RNN-based deep learning framework.

The proposed multi-scale RNN-based deep learning framework based on the bottleneck features [5] and the background property of transfer learning [6] can reflect both the reactivity and the accuracy with considering these two concatenated embedding results.

III. IMPLEMENTATION

A. Output Load input data

In this study, we proposed the RNN-based model with Output Load data collected hourly, every day for six months from Kookmin University, Korea. The data consists includes the Load Active Power and Periphery Temperature. Input data includes the night time when most public activity is finish, and the output load becomes very small. The input data are well-defined and fully collected, make sure that there is no missing and redundant data.

B. Prototype Implementation

The proposed model has the multi-scale RNN-based architecture with 200 RNN hidden units, 200-dimension bottleneck feature. The duration of short-term RNN model is set to 6 hours and long-term RNN is set to 24 × 7 h (a week). The duration of input data and the number of RNN hidden units are set identically. The bottleneck features of the multiscale RNN-based model are concatenated with 32 and 168-ratio and they are extracted from short-term and long-term RNN models, respectively. The weights of all RNN cells are initialized via Xavier [7] and are applied RNNDROP [8] with 0.7 keeping probability. The activation function for Combined features layer is Leaky ReLU [9], and mean square error (MSE) is used for loss function. The adopted iterative optimization algorithm is stochastic gradient descent (SGD) with the minibatch size of 168 using ADAM solver [10] and finally learning rate is set to 0.0005. The training was with 2,000 epochs for the proposed multi-scale RNN-based model.

C. Model evaluation metric

Error evaluation metrics were proposed to evaluate the overall result of Load Forecasting Scheme:

Root mean square error (*RMSE*): measures the difference between the forecasting values and the observed value. *RMSE* represents the square root of the differences between forecasting values and observed values:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(P_f - P_o)^2}$$

where N is the number of observations; P_f is the forecast value; P_o is the actual value.

Mean Absolute Error (*MAE*): *MAE* is the average of absolute errors. It can better reflect the accuracy of the forecasting value error, and is defined:

$$MAE = \frac{\sum_{i=1}^{N}|P_f - P_o|}{N}$$

The proposed evaluations are *RMSE* and *MAE*, which are suitable for evaluating the short term prediction, such as hourly loading forecasting.

D. Evaluation and Comparison

Figure 3 shows the result of multi-scale RNN-based deep learning with the training of 2000 epochs. The blue colored line denotes the real Output Load data and the blue one denotes the forecast values. The forecast values are closed to the real value with the same output pattern and peak time in the specific days. However, the peaks of prediction value are much lower than the real value, which is inappropriate for Output Load forecasting. Figure 4 shows the change of MSE and RMSE score for the test set according to the training iteration step of each model.

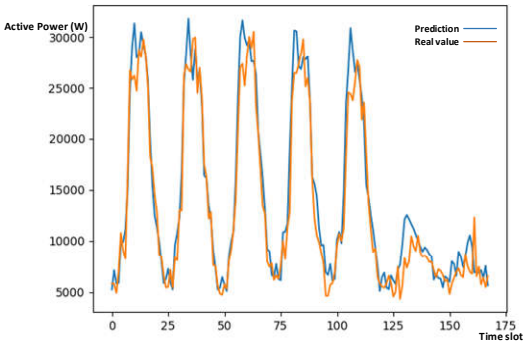


Figure 3: Output Load Forecasting with 2000 Epoch Training

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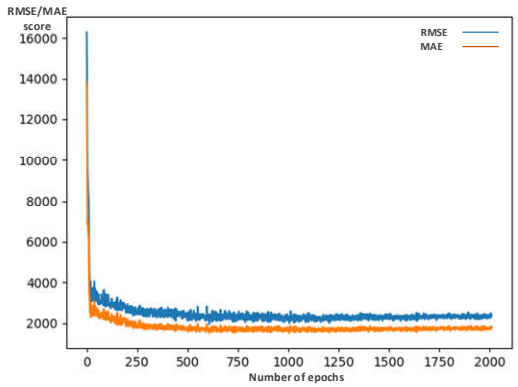


Figure 4: RMSE/MAE scores of the Output Load Forecasting

IV. CONCLUSION

Output Load Prediction is growing rapidly, becomes one of the most important tools for energy management nowadays. With the increasing of the requirement for accuracy of Output Load Prediction, RNN method becoming a robust solution with many advantages. In this paper, we proposed a RNN scheme for the Output Load Forecasting, as it applied the characteristics of a continuous input data approach. The proposed multi-scale RNN-based deep learning can reflect both the reactivity and the accuracy with high accuracy and shows the potential to be a robust solution for the Output Load Forecasting problems.

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REFERENCES

[1] Y. Wang, Q. Xia and C. Kang, "Secondary Forecasting Based on Deviation Analysis for Short-Term Load Forecasting," IEEE Transactions on Power Systems, pp. 500 - 507, May 2011.

[2] J. Vermaak and E. Botha, "Recurrent neural networks for short-term load forecasting," IEEE Transactions on Power Systems, pp. 126 - 132, Feb. 1998.

[3] J. B. C. E. Zachary C. Lipton, "A Critical Review of Recurrent Neural Networks for Sequence Learning," arXiv, 29 May 2015.

[4] D. Kim, S.-W. Hwang and J. Kim, "Very Short-Term Photovoltaic Power Generation Forecasting with Convolutional Neural Networks," in 2018 International Conference on Information and Communication Technology Convergence, Jeju, South Korea, 17-19 Oct. 2018.

[5] D. Yu and M. Seltzer, "Improved Bottleneck Features Using Pretrained Deep Neural Networks," in 12th Annual Conference of the International Speech Communication Association, Florence, Italy, August 2011.

[6] S. J. Pan and Q. Yang, "A Survey on Transfer Learning," IEEE Transactions on Knowledge and Data Engineering, pp. 1345 - 1359, 16 October 2009.

[7] Y. B. Xavier Glorot, "Understanding the difficulty of training deep feedforward neural networks," in Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics, January 2010.

[8] T. Moon, H. Choi, H. Lee and I. Song, "RNNDROP: A novel dropout for RNNS in ASR," in 2015 IEEE Workshop on Automatic Speech Recognition and Understanding, Scottsdale, AZ, USA, Dec. 2015.

[9] N. W. T. C. M. L. Bing Xu, "Empirical Evaluation of Rectified Activations in Convolutional Network," arXiv, Nov 2015.

[10] J. B. Diederik P. Kingma, "Adam: A Method for Stochastic Optimization," in 3rd International Conference for Learning Representations, San Diego, 2015.

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